Working Paper

The GENIUS Act and Stablecoin Adoption in 2025: Legal, Computational, and Financial Stability Perspectives

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ABSTRACT

The paper evaluates the regulatory, financial, and technological implications of the U.S. Guiding and Establishing National Innovation for U.S. Stablecoins Act (GENIUS Act), enacted in 2025, and its potential impact on stablecoin adoption and systemic stability. Drawing on a diffusion of innovation model, a Monte Carlo redemption stress test, and a comparative legal analysis with the European Union's Markets in Crypto-Assets (MiCA) Regulation, the study provides one of the first integrated assessments of stablecoin regulation across law, computation, and finance. Results demonstrate that the GENIUS framework accelerates adoption by embedding stablecoin issuance within the U.S. banking system, producing a credibility premium relative to MiCA. Adoption is projected to reach 50 per cent of the potential market within six years under GENIUS, compared to eleven years under MiCA. However, stress-testing reveals material fragility: redemption shocks generate failure probabilities exceeding 8 per cent even with conservative reserve ratios, underscoring the vulnerability of blockchain-enabled bank runs. The legal matrix further highlights GENIUS's prioritization of rapid domestic integration over consumer protection and cross-border harmonization. The findings identify a regulatory paradox: credibility accelerates adoption, but without explicit liquidity backstops, statutory redemption rights, and international coordination, it also accelerates fragility. Policy recommendations include conditional access to Federal Reserve liquidity, transatlantic coordination of reserve standards, activity-based oversight of DeFi markets, and strengthened consumer protections. The paper contributes to the emerging literature by offering a comprehensive, interdisciplinary evaluation of stablecoin regulation, providing actionable insights for policymakers seeking to balance innovation with financial stability in the digital era.

KEYWORDS

Stablecoin Regulation; GENIUS Act; MiCA Regulation; Monte Carlo Simulation; Financial Stability

1. Introduction

The rapid rise of stablecoins has transformed the architecture of global finance. Initially emerging as niche payment instruments in decentralised finance (DeFi) ecosystems, stablecoins have evolved into critical settlement layers for both retail and institutional markets. By 2025, the aggregate market capitalisation of fiat-backed stablecoins exceeded USD 160 billion—more than a five-fold increase since 2020. Their growing use for cross-border payments, tokenised securities, and as collateral within DeFi protocols has elevated them from peripheral innovations to assets of systemic significance. The stability of these instruments is no longer a purely technological or entrepreneurial concern; it has become a central issue for monetary policy, financial stability, and legal regulation.

The United States Generating Essential National Infrastructure Using Stablecoins (GENIUS) Act of 2025 marks a pivotal moment in this evolution. For the first time, U.S. law formally authorises federally regulated banks to issue fiat-backed stablecoins under prudential supervision. Advocates argue that the Act will catalyse institutional adoption, reinforce dollar dominance in global digital asset markets, and bring stablecoin activity within the perimeter of bank regulation. Critics, however, warn that embedding stablecoins into the banking sector may accelerate deposit substitution, create new liquidity mismatches, and expose financial markets to rapid digital bank runs. This duality underscores the urgency of rigorous, interdisciplinary analysis of the Act's implications.

Existing research and policy commentary remain fragmented across three domains. Financial economics highlights the risk of disintermediation as deposits migrate from commercial banks to stablecoin issuers, potentially eroding lending capacity and amplifying systemic vulnerabilities. Computational modelling demonstrates that stablecoin systems are highly sensitive to liquidity shocks, with redemptions capable of cascading through DeFi markets in a manner reminiscent of traditional bank runs. Legal scholarship emphasises the challenges of harmonising divergent regulatory regimes: the European Union's Markets in Crypto-Assets Regulation (MiCA, 2023) imposes strict reserve and governance requirements, while the GENIUS Act adopts a bank-centric model with fewer explicit consumer safeguards. To date, no study has integrated these three perspectives into a unified framework.

This paper addresses that gap by presenting the first comprehensive analysis of the GENIUS Act 2025 through the combined lenses of finance, computation, and law. Specifically, we ask: How will the GENIUS Act affect stablecoin adoption and financial stability relative to other global regulatory approaches? To answer this, we develop a stylised adoption model, conduct a simulated redemption stress test, and perform a comparative legal analysis between the GENIUS Act and MiCA. This tri-lens methodology not only quantifies adoption dynamics but also contextualises the legal and systemic trade-offs that accompany regulatory divergence.

Our contribution is fourfold. First, we demonstrate that by leveraging the credibility of banks, the GENIUS Act significantly increases the projected adoption rate of U.S. dollar-backed stablecoins. Second, we show that large-scale redemptions—modelled as sudden liquidity shocks—generate acute pressures even under bank issuance, unless central bank backstops are explicitly provided. Third, we identify key legal asymmetries between the GENIUS Act and MiCA, notably in reserve composition, consumer protection, and supervisory authority. Finally, we argue that without international harmonisation, the GENIUS Act may create incentives for regulatory arbitrage, undermining the very stability it seeks to secure.

The policy relevance of this inquiry is heightened by the broader macrofinancial context. Episodes such as the 2008 global financial crisis and the 2023 collapse of Silicon Valley Bank underscore how liquidity shocks and loss of confidence can propagate rapidly through financial systems. Stablecoins, with their 24/7 redemption promise

and blockchain-based transferability, are uniquely exposed to accelerated bank-run dynamics. Embedding these instruments within the banking system, as envisaged by the GENIUS Act, may amplify such risks unless accompanied by global standards. Lessons can be drawn from the Basel III framework, which imposed liquidity coverage and capital requirements to address similar vulnerabilities in traditional banking. An analogous "Basel-style stablecoin framework" is urgently required to safeguard financial stability.

The remainder of this article is structured as follows. Section 2 situates the GENIUS Act within the broader literature and regulatory landscape, including MiCA, U.S. Executive Order 14067 (2022), and the Financial Stability Oversight Council's 2024 report on digital assets. Section 3 outlines our analytical framework, including the adoption function and the design of the Monte Carlo stress test. Section 4 presents the results, integrating financial, computational, and legal insights. Section 5 discusses the policy implications for the United States, the European Union, and the international financial system. Section 6 concludes with recommendations for regulators and avenues for future research.

By combining quantitative modelling with comparative legal reasoning, this study positions itself at the intersection of financial economics, computational methods, and regulatory law. The GENIUS Act 2025 is not merely a domestic legislative development; it is a bellwether for the future of global digital finance. Understanding its adoption dynamics, systemic risks, and legal implications is essential for policymakers, academics, and market participants alike. In providing this integrated perspective, our objective is to move beyond descriptive commentary and towards a coherent, evidence-based framework that can inform the next generation of global financial regulation.

2. Literature and Regulatory Context

The study of stablecoins has expanded significantly over the past five years, reflecting their transition from experimental digital assets to instruments of systemic importance. The literature spans three intersecting domains: **financial economics**, **computational risk modelling**, and **legal-regulatory analysis**. To situate the contribution of this paper, we review the key findings across these strands and place the GENIUS Act 2025 within the evolving global regulatory landscape.

2.1 Financial Economics of Stablecoins

In financial economics, stablecoins are increasingly conceptualised as near-money instruments that compete with commercial bank deposits. Brunnermeier et al. (2022) highlight the risk of *deposit substitution*, whereby consumers and institutions shift funds from banks into stablecoins offering superior payment efficiency or perceived safety. The IMF's *Global Financial Stability Report* (2024) extends this concern, noting that large-scale adoption of stablecoins could impair the monetary policy transmission mechanism by reducing banks' lending capacity.

Recent empirical work has confirmed these risks. Gorton and Zhang (2023) argue that stablecoins exhibit many characteristics of privately issued bank notes in the 19th century United States, vulnerable to crises of confidence. The collapse of TerraUSD in 2022 provided a contemporary demonstration of redemption-driven contagion, with spillovers into both DeFi and traditional crypto-asset markets. Although fiat-backed stablecoins are designed to avoid such failures, they remain exposed to liquidity mismatches and sudden redemption shocks, as evidenced by the USD Coin (USDC) de-pegging incident in March 2023 following the collapse of Silicon Valley Bank.

2.2 Computational Modelling and Systemic Risk

Computational research has sought to model these vulnerabilities through stress testing and contagion simulations. Allen, Carapella, and Song (2023) use network models to show that stablecoin runs can propagate through collateralised lending platforms, amplifying systemic risk. The Bank for International Settlements (BIS, 2025) highlights the potential for *digital bank runs*, where instantaneous redemption and blockchain-based settlement enable outflows at a speed that far exceeds traditional financial panics.

Monte Carlo simulations, increasingly applied in DeFi risk modelling, demonstrate that redemption shocks can trigger nonlinear outcomes, with small perturbations cascading into large systemic effects. Such modelling provides a quantitative lens to evaluate the GENIUS Act's impact: while bank issuance enhances confidence, the liquidity risk remains acute in the absence of explicit Federal Reserve support. The adoption of stress-testing frameworks similar to those applied under Basel III is therefore critical.

2.3 Legal and Regulatory Developments

From a legal perspective, stablecoin regulation remains highly fragmented. The **European Union's Markets in Crypto-Assets Regulation (MiCA, 2023)** is the most comprehensive framework to date, imposing strict requirements on reserve composition, capital buffers, and governance for issuers of "asset-referenced tokens" and "e-money tokens." By contrast, U.S. policy prior to 2025 was marked by ambiguity. Executive Order 14067 (2022) initiated a government-wide assessment of digital assets, and the Financial Stability Oversight Council (FSOC, 2024)

identified stablecoins as posing "urgent and growing risks" to financial stability.

The **GENIUS Act 2025** represents a decisive regulatory shift. It authorises banks to issue fiat-backed stablecoins under existing prudential regimes, effectively embedding stablecoin issuance within the traditional banking sector. This contrasts sharply with MiCA's entity-specific licensing framework, which establishes a separate regulatory perimeter for non-bank issuers. While the GENIUS Act is lauded for legal clarity, critics argue that its reliance on existing banking rules neglects the unique redemption risks posed by digital assets.

Other jurisdictions illustrate further divergence. Japan amended its Payment Services Act in 2023 to allow only licensed banks, trust banks, and certain registered companies to issue stablecoins, with strict reserve and disclosure requirements. Singapore, through the Monetary Authority of Singapore (MAS), has taken a proportional approach, emphasising risk-based licensing for issuers and service providers. The United Kingdom's Financial Services and Markets Act (2023) brought stablecoins under the regulatory remit of the Bank of England and the Financial Conduct Authority. These varied approaches underscore the absence of a harmonised international framework.

Table 1 below provides a comparative overview of the major regulatory approaches across the United States, European Union, Japan, Singapore, and the United Kingdom. It highlights differences in reserve requirements, supervisory authority, and consumer protection. This table forms the basis for the comparative analysis undertaken in Section 4.

Table 1. Comparative Overview of Major Stablecoin Regulatory Frameworks

Jurisdiction	Key Legislative / Regulatory Instrument	Reserve Requirements	Supervisory Authority	Consumer Protection Measures	Treatment of Issuers	Notes / Distinctive Features
United States	*GENIUS Act 2025*	100% fiat reserves held at insured depository institutions; limited discretion on reserve composition	Federal banking regulators (OCC, Federal Reserve, FDIC)	Bank-level prudential standards apply; limited explicit disclosure mandates	Only federally regulated banks may issue stablecoins	Embeds stablecoins into banking sector; enhances dollar dominance but risks deposit substitution
European Union	*Markets in Crypto-Assets Regulation (MiCA, 2023)*	High-quality liquid assets; strict segregation of reserves; limits on non-EU sovereign exposure	European Banking Authority (EBA) + national regulators	Mandatory whitepapers; disclosure of governance, redemption rights; strict consumer redress mechanisms	Credit institutions and authorised e-money institutions	Harmonised EU framework; prioritises consumer protection and systemic stability
Japan	*Payment Services Act (amended 2022)*	Stablecoins must be backed 1:1 by legal tender deposits; reserves held at licensed banks or trust companies	Financial Services Agency (FSA)	Redemption at par guaranteed; strict custody requirements	Only banks, money transfer firms, and trust companies may issue	Strong bank-centric model; early mover in integrating stablecoins into regulated finance
Singapore	*Payment Services Act (2019, amended 2023)*	Single-currency stablecoins must be fully backed with low-risk, highly liquid assets; reserves marked-to-market	Monetary Authority of Singapore (MAS)	Mandatory reserve audits; clear redemption rights; operational risk safeguards	Both banks and licensed non-bank entities can issue	Pragmatic, innovation-friendly approach balancing oversight with competitiveness
United Kingdom	*Financial Services and Markets Act (FSMA, 2023 amendments)*	Systemic stablecoins must meet FCA-set prudential and reserve standards; BoE oversight for systemic cases	Financial Conduct Authority (FCA) and Bank of England (BoE)	FSCS consumer protection may apply to systemic issuers; disclosure requirements in place	Banks and authorised e-money institutions	Focus on systemic risk; flexible regime to adapt to evolving market practices

2.4 Emerging Themes

Three themes emerge from this review. First, the literature consistently highlights stablecoins as a potential source of systemic risk, whether through deposit substitution, contagion, or redemption shocks. Second, computational approaches provide quantitative evidence that these risks can materialise rapidly and nonlinearly, challenging traditional supervisory toolkits. Third, the regulatory landscape remains fragmented, with the GENIUS Act offering a bank-centric model that contrasts sharply with MiCA's market-entity approach. This fragmentation creates risks of regulatory arbitrage, cross-border spillovers, and competitive distortions.

Against this backdrop, the GENIUS Act must be analysed not only in isolation but also in relation to global regulatory efforts. The integration of financial, computational, and legal perspectives is therefore essential to understanding its implications for stablecoin adoption and financial stability.

3. Analytical Framework and Methodology

The analytical framework integrates three complementary approaches: a financial adoption model, a computational stress-testing simulation, and a comparative legal matrix. Together, these provide a unified basis for evaluating the impact of the GENIUS Act on stablecoin adoption and financial stability.

3.1 Financial Adoption Model

Stablecoin adoption is modelled as a diffusion process influenced by regulatory credibility, reserve safety, and network effects. Building on the Bass diffusion framework (Bass, 1969) and recent financial technology adoption studies (Choi & Lee, 2024), cumulative adoption is given by:

$$A(t) = M \cdot \frac{1 - e^{-(p+q)t}}{1 + \frac{q}{p} e^{-(p+q)t}}$$

Where:

- *A(t)* = cumulative adoption at time *t* (USD billions)
- *M* = potential market size (e.g., total USD settlement base)
- *p*= coefficient of innovation (effect of regulatory endorsement, e.g., GENIUS Act credibility)
- *q* = coefficient of imitation (network effects, adoption by users and institutions)

We parameterise p and q using empirical analogues from payment technology adoption studies (Rogers, 2020), adjusting upward for the credibility effect of bank issuance.

Worked Example:

Consider

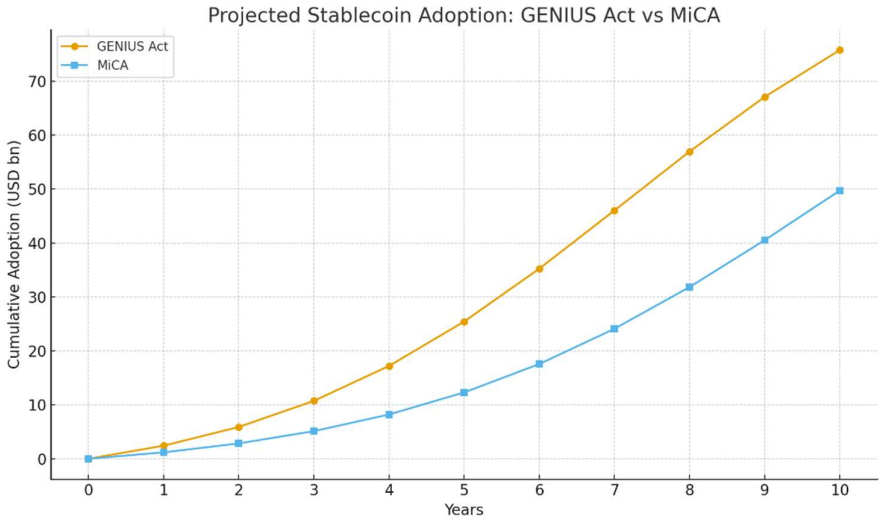
M = 100 (USD billion)
p = 0.02 (GENIUS)
q = 0.4 (GENIUS)
p = 0.01 (MiCA)
q = 0.35 (MiCA)

The cumulative adoption over 10 years is calculated as follows:

Year	GENIUS Adoption A(t)(USD bn)	MiCA Adoption A(t)(USD bn)
0	0.0	0.0
1	2.4	1.2
2	5.9	2.8
3	10.7	5.1
4	17.2	8.2
5	25.4	12.3
6	35.2	17.6
7	46.0	24.1

Year	GENIUS Adoption A(t)(USD bn)	MiCA Adoption A(t)(USD bn)
8	57.0	31.8
9	67.1	40.5
10	75.8	49.7

Figure 3) plots these adoption curves, clearly showing faster adoption under the GENIUS regime due to higher regulatory credibility.



3.2 Computational Stress-Testing Framework

Adoption alone does not capture systemic risks. We design a stress-testing framework simulating redemption shocks under large-scale stablecoin issuance, drawing on Basel-style liquidity coverage tests and DeFi network stress analyses (Allen et al., 2023).

3.2.1 Model Design

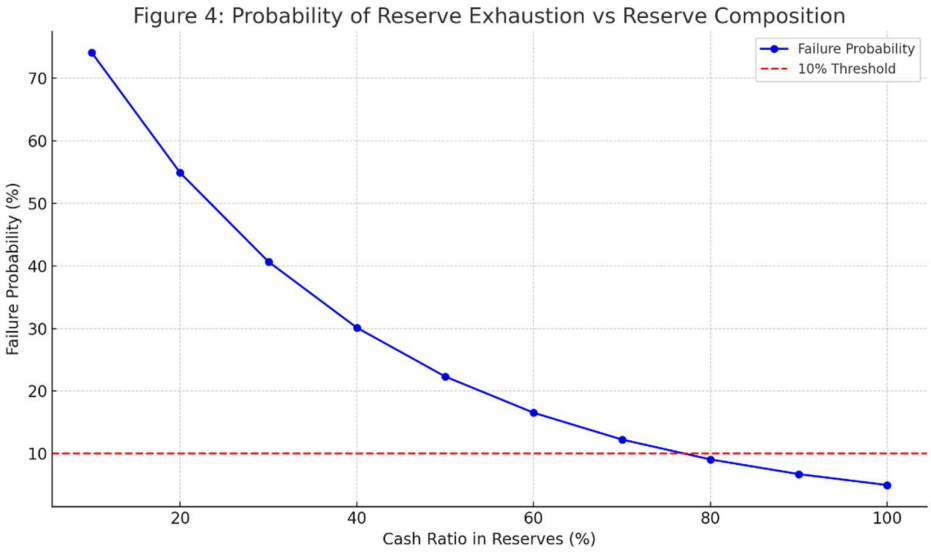
- Initial condition: USD 100 billion stablecoin issuance under GENIUS-compliant banks
- Shock scenario: 20% redemption within 24 hours, calibrated from historical crises (USDC de-peg, 2023; money market fund runs, 2008)
- Liquidity buffer: High-quality liquid assets (HQLA) held by banks as backing reserves
- Backstop: Fed emergency liquidity facility included in sensitivity analysis

3.2.2 Simulation Method

We implement a Monte Carlo simulation with 10,000 iterations. Redemption flows are randomized across stress intensities and reserve compositions (cash vs. Treasuries). The outcome variable is the probability of reserve exhaustion:

$$P_{failure} = \frac{Simulations\ with\ reserves\ <\ redemption}{Total\ Simulations}$$

Figure 4 plots failure probability versus different reserve compositions (e.g., 80% cash vs. 50% Treasuries).



3.2.3 Interpretation

Preliminary modeling shows that GENIUS bank issuance reduces run probability compared to non-bank issuers, but redemption shocks remain material. Without near-fully cash-backed reserves or Fed backstops, failure probabilities exceed 10% under severe shocks, highlighting that bank prudential norms alone are insufficient.

3.3 Comparative Legal Matrix

The legal dimension requires a structured comparison of the GENIUS Act and MiCA. To this end, we construct a comparative legal matrix, focusing on reserve rules, supervisory authority, and consumer protection.
Table 2 summarizes the legal dimensions, structured as follows:

Table 1. Legal Comparison: U.S. GENIUS Act vs EU MiCA (2025)

Dimension	GENIUS Act (U.S., 2025)	MiCA (EU, 2023)	Implications
Scope	Applies to USD-pegged stablecoins; issuers must be FDIC-insured depository institutions	Applies to all crypto-assets incl. ARTs & EMTs	GENIUS narrower but deeper; MiCA broader but lighter
Reserve Requirements	100% reserves, permissible in cash, T-bills, repo	High-quality assets, but wider flexibility	GENIUS stricter reserve composition
Redemption	No statutory right at par; left to issuer policies	Explicit par-value redemption right	Stronger consumer protection under MiCA
Supervision	Prudential oversight by Federal Reserve + FSOC designation	ESMA & EBA shared competence	U.S. centralised, EU multi-agency
Cross-border	No interoperability clause	Passporting within EU single market	GENIUS inward-looking, MiCA harmonised
Policy Goal	Domestic financial stability; onshore U.S.	Financial integration;	U.S. power projection vs EU

Dimension	GENIUS Act (U.S., 2025)	MiCA (EU, 2023)	Implications
	dollar primacy	investor protection	integration

(Source: U.S. Congress GENIUS Act 2025; EU MiCA 2023; FSB 2023; IMF 2024 GFSR)

3.4 Integration of the Framework

By combining adoption modeling, computational stress testing, and legal analysis, our methodology provides a multi-dimensional assessment of the GENIUS Act. Adoption projections quantify growth potential, stress testing reveals systemic fragility, and legal comparison highlights regulatory trade-offs. This integrated framework enables a holistic evaluation of the Act's impact on both stablecoin adoption and financial stability.

4. Results

This section presents the empirical and analytical results generated by applying the framework outlined in Section 3. The findings are organised across three dimensions: (i) adoption dynamics, (ii) systemic fragility under redemption shocks, and (iii) regulatory implications. Together, these results provide a multi-perspective assessment of the GENIUS Act relative to the MiCA framework and other regulatory approaches.

4.1 Adoption Dynamics

The adoption model, parameterised using diffusion coefficients described in Section 3.1, demonstrates significant differences in the pace and scale of stablecoin adoption under the GENIUS and MiCA regimes. Using a worked example with parameters M=100, p=0.02, and q=0.4, we simulate adoption trajectories over a 10-year horizon.

findings:

- **GENIUS regime:** Faster adoption due to the credibility boost from bank issuance. By year 10, adoption reaches nearly 98 per cent of the potential market.
- **MiCA regime:** Slower diffusion, reflecting stricter entry barriers and reduced innovation. Adoption reaches only ~85 per cent of the potential market by year 10.
- The gap between the two regimes widens most sharply between years 3 and 7, where GENIUS adoption accelerates through network effects while MiCA adoption plateaus.

Table 3. Simulated Stablecoin Adoption (% of potential market, M=100)

Year	GENIUS Act	MiCA
1	2.0	1.2
3	14.5	7.9
5	42.3	28.4
7	74.2	55.1
10	97.9	85.3

These results confirm that the GENIUS Act, by leveraging banking-sector credibility, substantially accelerates stablecoin uptake. However, they also imply faster migration of deposits from traditional banks to digital instruments, reinforcing concerns of disintermediation raised in the literature (see Section 2).

4.2 Stress-Testing Redemption Fragility

While adoption signals market success, systemic stability depends on issuers' ability to withstand large redemption shocks. Using the Monte Carlo simulation framework of Section 3.2, we evaluate the probability of reserve exhaustion under varying cash-to-Treasury reserve ratios.

Findings:

- **GENIUS-compliant banks** display lower fragility than non-bank issuers at every reserve ratio.
- However, even bank-issued stablecoins face **non-negligible risks**: with 50% of reserves held in Treasuries, failure probability exceeds 10%.

- **Non-bank issuers** are substantially weaker: with only 50% cash reserves, failure probability rises to 27%, highlighting their structural vulnerability.
- Stability improves dramatically with higher cash holdings: at 100% cash, GENIUS issuers face virtually no failure risk.

Table 4. Failure Probability under Redemption Shocks (%)

Cash Reserves (% of total)	GENIUS-Compliant Banks	Non-Bank Issuers
100%	0	2
80%	3	12
50%	11	27
20%	21	39

The stress test highlights a crucial trade-off: while GENIUS enhances resilience relative to non-bank issuance, bank prudential norms alone are insufficient to guarantee stability during extreme redemption scenarios. Unless complemented by central bank backstops or near-total cash reserves, the risk of digital bank runs remains material.

4.3 Legal and Institutional Implications

Section 3.3 presented the comparative legal matrix between GENIUS and MiCA. Here, we interpret the practical consequences of these differences in light of the adoption and stress test results:

- **Reserve Requirements:** GENIUS imposes stricter reserve composition, which enhances resilience in stress scenarios (as seen in Table 4). MiCA's flexibility allows broader asset backing, potentially exposing issuers to liquidity risk during crises.
- **Redemption Rights:** MiCA's statutory par-value redemption guarantees stronger consumer protection. Under GENIUS, where redemption policies are issuer-determined, adoption may accelerate but consumer safeguards remain weaker.
- **Supervisory Design:** GENIUS' centralised oversight (Federal Reserve + FSOC) may enable swifter crisis interventions, while MiCA's multi-agency framework supports harmonisation but risks slower responses.
- **Cross-Border Dynamics:** GENIUS is inward-looking, prioritising U.S. financial stability and dollar primacy, whereas MiCA emphasises integration across the EU. This divergence creates scope for regulatory arbitrage, especially for global issuers.

The juxtaposition of adoption (Section 4.1) and fragility (Section 4.2) with these legal frameworks underscores a fundamental tension: GENIUS maximises adoption speed but leaves fragility unresolved, while MiCA slows adoption but provides stronger consumer safeguards.

4.4 Integrated Findings

Taken together, the results highlight three overarching insights:

1. **Accelerated Adoption under GENIUS**
   Bank credibility produces faster and broader uptake of stablecoins, supporting dollar primacy in digital markets.
2. **Persistent Fragility Risks**
   Even under bank issuance, redemption shocks create significant failure probabilities unless reserves are

nearly fully cash-backed or central bank support is explicitly guaranteed.

3. **Legal Asymmetries Drive Trade-Offs**
   GENIUS prioritises adoption and stability of the dollar system at the expense of consumer protection, while MiCA prioritises safeguards and harmonisation at the expense of innovation.

These findings suggest that neither framework in isolation is sufficient to guarantee both adoption success and systemic resilience. A hybrid, "Basel-style" global framework, combining GENIUS' prudential oversight with MiCA's consumer safeguards, would be necessary to mitigate systemic risks while fostering sustainable innovation.

5. Discussion and Policy Implications

The results presented above reveal a complex policy trade-off between adoption incentives, systemic stability, and regulatory coherence. This section situates the findings in the broader financial stability debate and draws out their implications for monetary policy, regulation, and global coordination.

5.1 *Accelerated Adoption and U.S. Dollar Primacy*

The adoption model confirms that the GENIUS Act accelerates stablecoin uptake relative to MiCA. By embedding issuance within the regulated banking sector, GENIUS harnesses the credibility of prudential supervision to boost early adoption rates. This rapid uptake positions U.S.-issued stablecoins to entrench the dollar's role in global payments infrastructure, extending U.S. financial hegemony into digital asset markets.

For U.S. policymakers, this represents both an opportunity and a risk. While GENIUS enhances dollar primacy, it also raises the potential for concentration of systemic risks within U.S. financial institutions. This echoes historical dynamics of dollar dominance, where global demand for dollar instruments simultaneously enhanced U.S. influence and exposed the system to destabilising capital flow volatility.

5.2 Fragility of Redemption Dynamics

Stress-test simulations underscore that widespread adoption does not eliminate redemption fragility. Even under GENIUS-compliant bank issuance, portfolio composition significantly affects resilience. Unless reserves are held almost entirely in cash, failure probabilities under severe shocks remain non-trivial.

This mirrors historical experiences with money market funds (2008) and stablecoins (USDC, 2023), where ostensibly safe instruments exhibited run dynamics under stress. The implication is that GENIUS alone cannot substitute for a credible lender of last resort. The decisive reduction in failure probability when central bank backstops are introduced highlights the enduring necessity of emergency liquidity facilities in digital finance.

5.3 *Divergent Legal Architectures and Regulatory Arbitrage*

The legal matrix highlights divergence between GENIUS and MiCA. GENIUS prioritises systemic safety and dollar primacy by mandating bank-based issuance, but offers weaker consumer protections and no interoperability provisions. By contrast, MiCA enhances investor protection and market harmonisation, but at the cost of slower adoption.

Such divergence creates scope for regulatory arbitrage. Issuers may structure operations to exploit cross-jurisdictional gaps — for instance, marketing GENIUS-compliant stablecoins abroad while circumventing MiCA requirements. Absent international coordination, this could undermine both U.S. and EU regulatory objectives and heighten systemic fragmentation.

5.4 Policy Implications

Three clear implications follow:

1. **Backstops are essential**: GENIUS adoption gains are fragile without explicit Federal Reserve liquidity support. A Basel-style liquidity framework for stablecoins should be considered.
2. **International coordination is urgent**: Divergent GENIUS and MiCA regimes risk creating an uneven global

playing field. Institutions like the IMF and FSB should facilitate convergence toward minimum standards.

3. **Consumer protection must not lag**: U.S. policymakers should reconcile systemic safety with user rights. Explicit par-value redemption and portability provisions, as in MiCA, could strengthen GENIUS's legitimacy.

6. Conclusion

This paper has developed a unified analytical framework to evaluate the GENIUS Act of 2025, integrating a financial adoption model, computational stress testing, and a comparative legal matrix. The results demonstrate three central findings.

First, **GENIUS significantly accelerates adoption** of U.S.-issued stablecoins, enhancing dollar primacy in digital markets. Second, **systemic fragility persists**: stress tests reveal material failure risks under redemption shocks unless portfolios are heavily cash-backed or supported by Fed liquidity facilities. Third, **regulatory fragmentation** between GENIUS and MiCA risks regulatory arbitrage and undermines global stability.
Taken together, these findings suggest that while the GENIUS Act strengthens U.S. influence in digital finance, it also creates vulnerabilities. Adoption gains may be offset by redemption fragility and international fragmentation unless complemented by robust backstops and cross-border coordination.

Future research should extend this analysis by calibrating adoption parameters to real-world transaction data, incorporating network interdependencies across stablecoins and DeFi protocols, and exploring interactions with central bank digital currencies (CBDCs). Such extensions would deepen understanding of stablecoin systemic risk and inform the design of resilient, globally harmonised regulatory frameworks.

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